

Ransera Price List Initial Analysis

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Introduction

The goal of this initial analysis is to examine the distribution of price for each of the broad categories of items available in Ransera. Within these categories there are additional subcategories which can be analyzed as well, but the analysis of these subcategories is outside the scope of this initial report.

Category
Animals
Armor
Battlefield_Gear
Clothing
Entertainment_Gear
Foodstuffs
Furniture
Land_and_Structures
Larceny_Gear
Lodging
Melee_Weaponry
Mounted_Related_Gear
Naval
Professions
Ranged_Weaponry
Religious_Gear
Slaves
Subterranean_Items
Tools_and_items
Trade_Goods
Wilderness_Exploration

Ransera Currency Use

The currency of Ransera is separated into 3 units of measure: Copper (cp), Silver (sp) and Gold (gp). In the setting of Ransera the cp is meant to be the benchmark for everyday use by characters. 100 cp is equal to 1 sp, and 10 sp is equal to 1 gp. With this relationship in mind we would expect the price list to reflect a greater number of items to cost n amount of cp or sp, however, the reality is the larger majority of the price

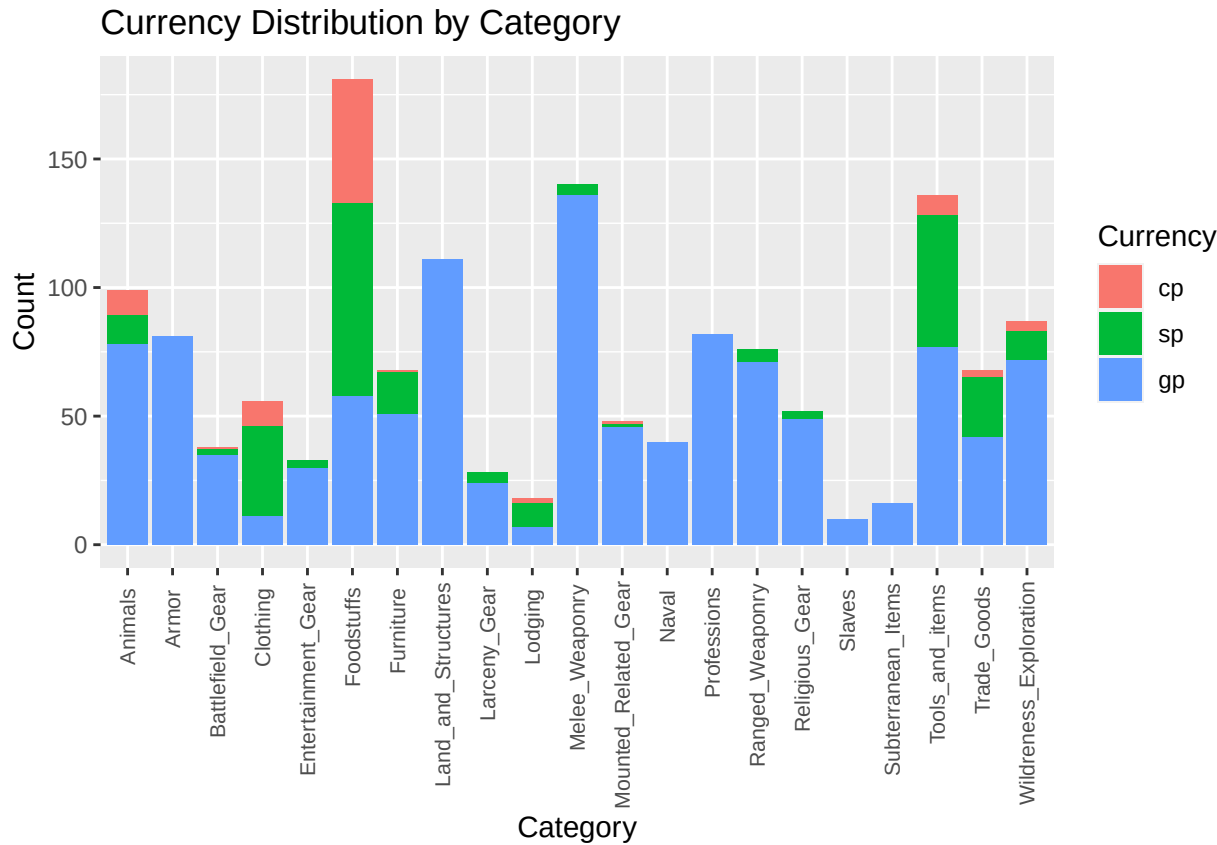
list utilizes the highest currency denomination. This suggests a possible imbalance in the current distribution of pricing across all categories.

Count of Currency Use
Number of times a currency is used

Currency	Count
cp	88
gp	1127
sp	253

Currency Distribution Across Price List

Looking at each of the categories with the figure below, we can see which categories have the most items in the price list, and the break down of how many of those items use cp, sp or gp as their currency. This view helps to visually highlight the abundance of items which cost gp (shown in blue) vs sp and cp (green and red respectively).



Minimum and Maximum Price by Category

We can also examine the highest and lowest price found within each category. Those Categories that have both a minimum and maximum price in gp are potentially priority targets for further examination and possible redistribution of pricing for their list items.

Minimum and Maximum Price per Category

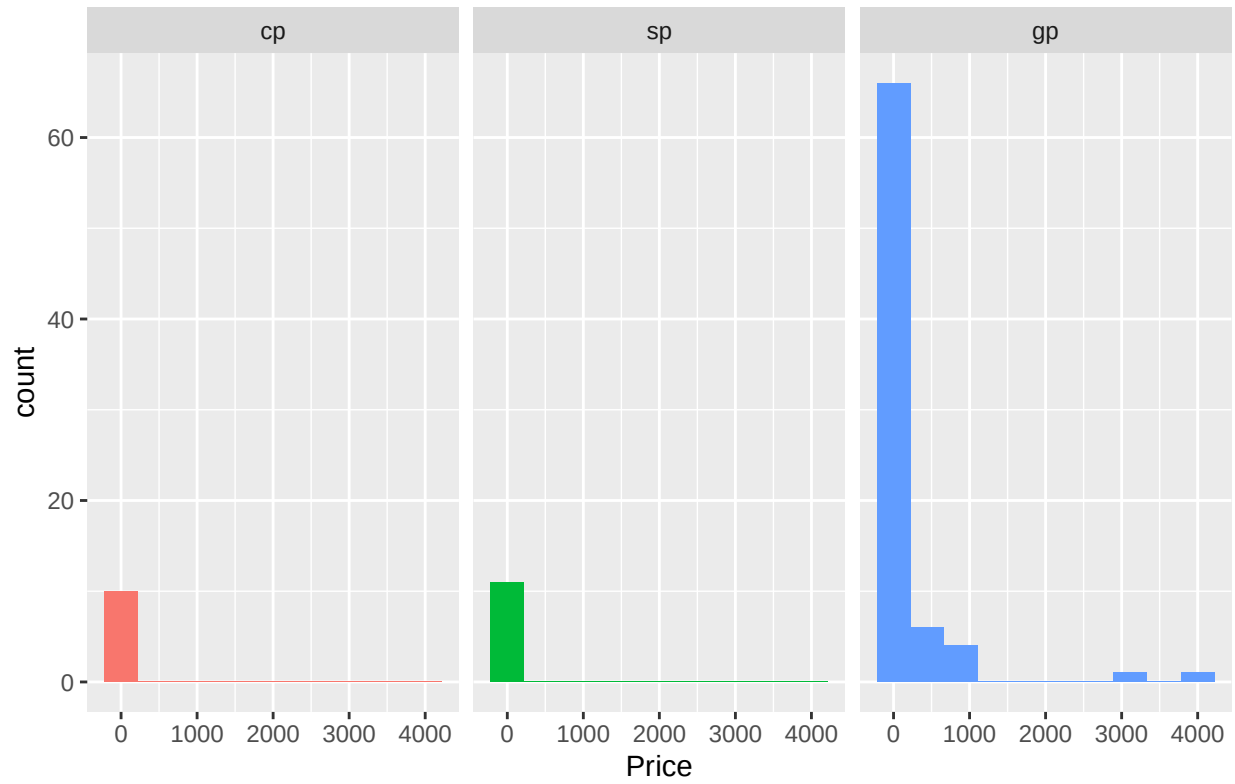
Min_Max	Price	Currency
Animals		
Max	4000	gp
Min	1	sp
Armor		
Max	5000	gp
Min	1	gp
Battlefield_Gear		
Max	500	gp
Min	1	cp
Clothing		
Max	45	gp
Min	1	cp
Entertainment_Gear		
Max	500	gp
Min	1	sp
Foodstuffs		
Max	100	gp
Min	1	cp
Furniture		
Max	75	gp
Min	1	gp
Land_and_Structures		
Max	50000	gp
Min	100	gp
Larceny_Gear		
Max	250	gp
Min	1	gp
Lodging		
Max	10	gp
Min	1	sp
Melee_Weaponry		

Max	300	gp
Min	1	sp
Mounted_Related_Gear		
Max	2400	gp
Min	1	gp
Naval		
Max	200000	gp
Min	1	gp
Ranged_Weaponry		
Max	15000	gp
Min	1	sp
Religious_Gear		
Max	2000	gp
Min	1	sp
Slaves		
Max	15000	gp
Min	1000	gp
Subterranean_Items		
Max	200	gp
Min	3	gp
Tools_and_items		
Max	150	gp
Min	1	sp
Trade_Goods		
Max	300	gp
Min	1	cp
Wildreness_Exploration		
Max	400	gp
Min	1	cp

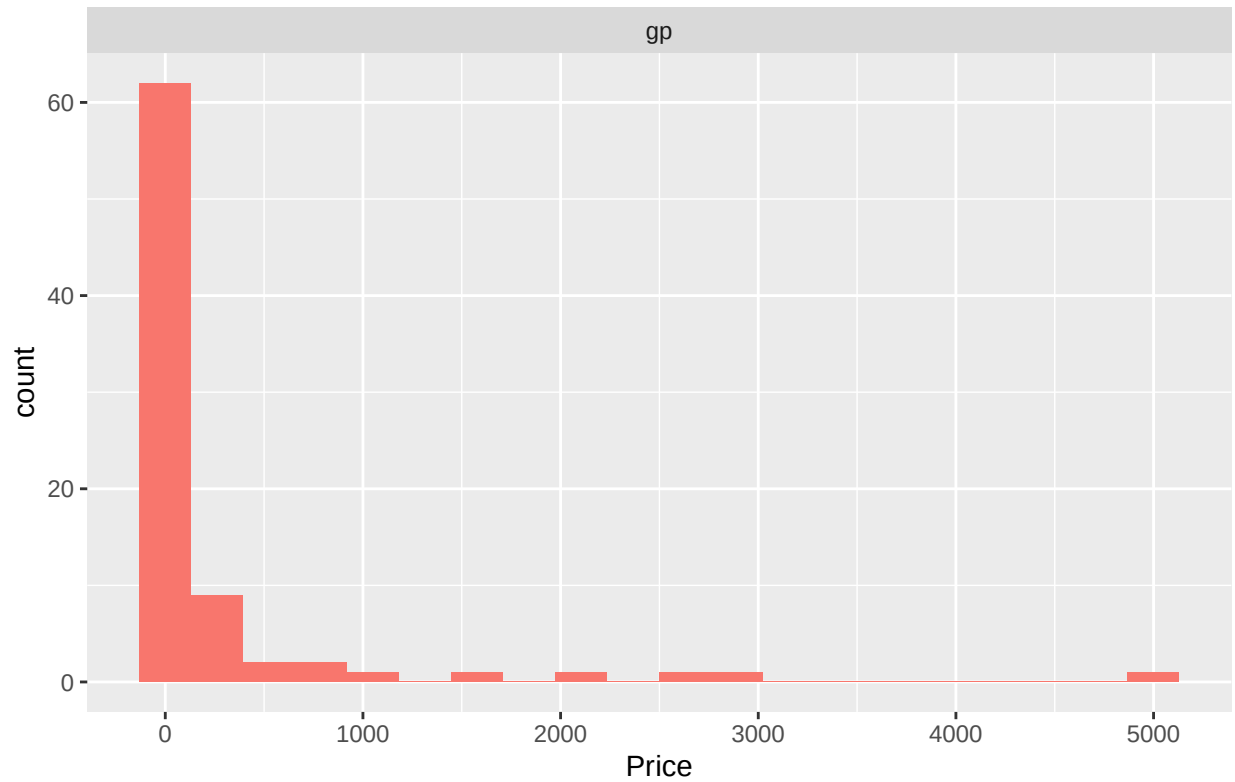
Price Distribution per Category

Finally a brief look at the distribution of pricing for each individual category. By studying these distributions it can be determined which categories should be prioritized for redistribution. Focus should be given to those Categories which possess prices above their currency shift range (i.e. 200 cp instead of 2 sp) and those which have a distribution that heavily favors gp over both cp and sp.

Animals Price Distribution

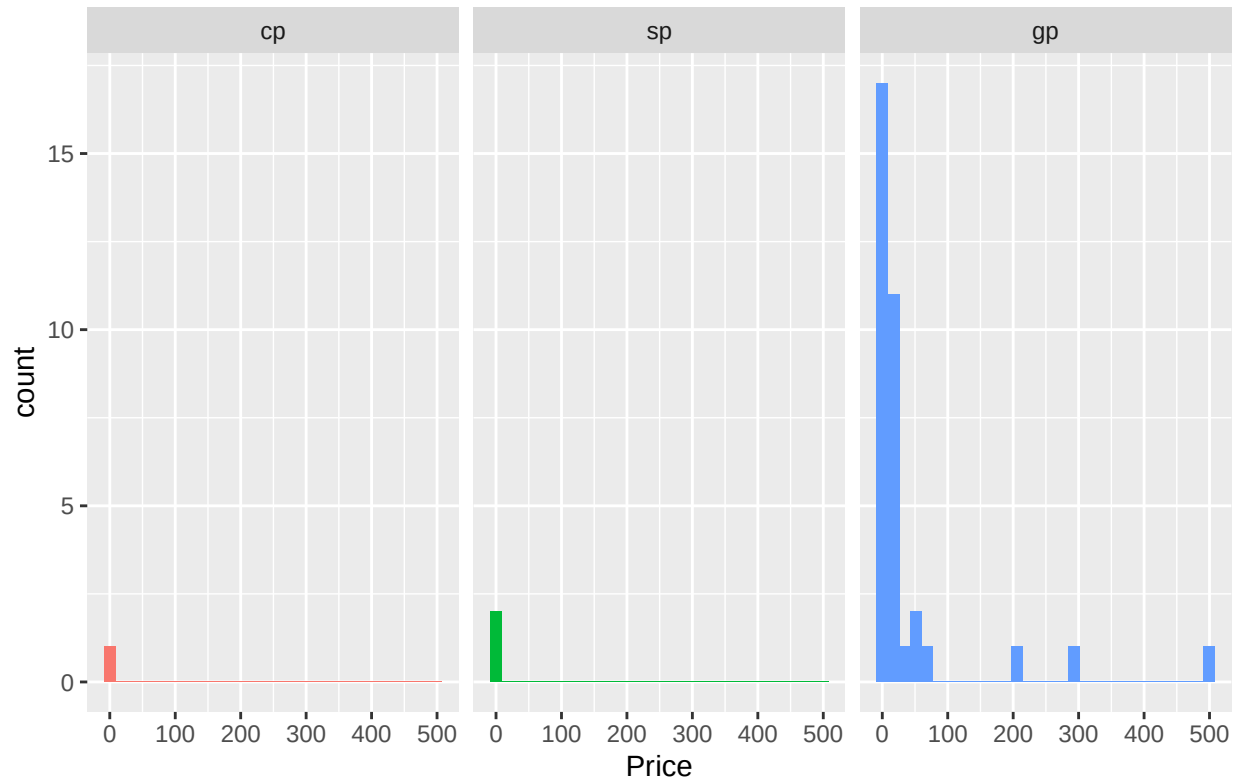


Armor Price Distribution



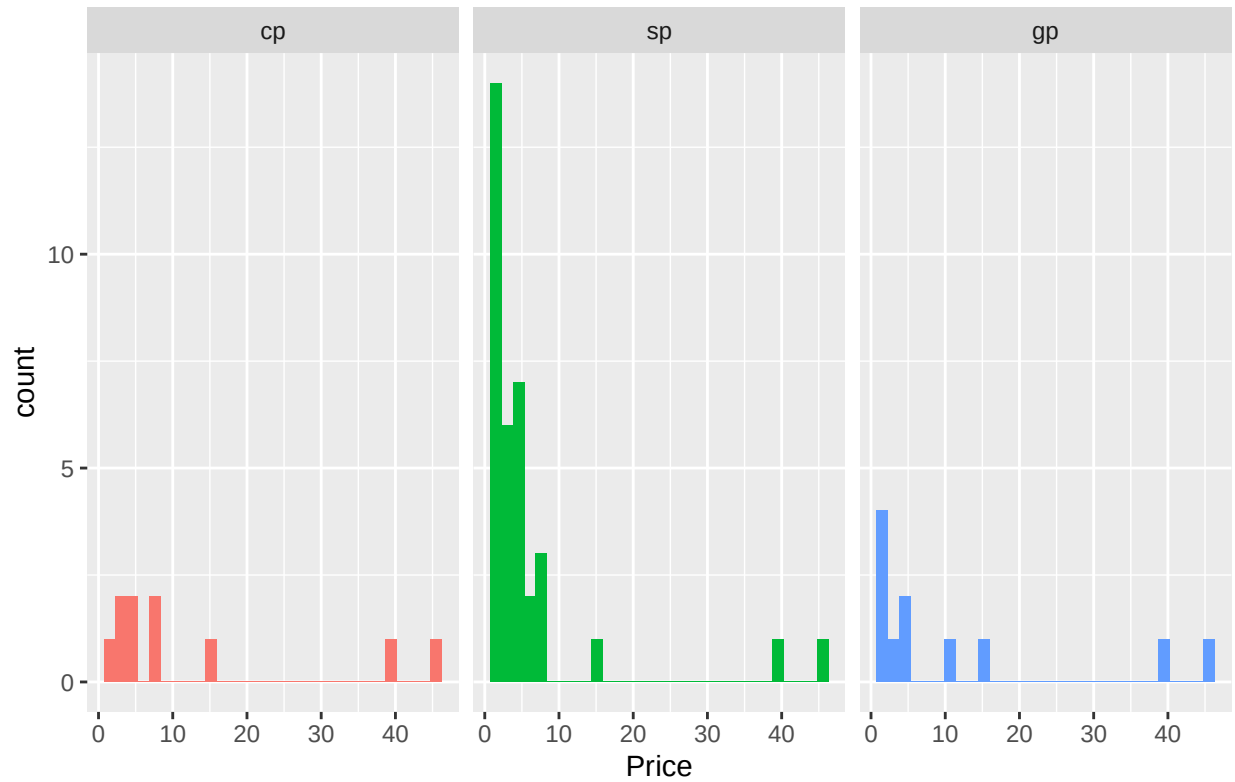
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## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Battlefield_Gear Price Distribution



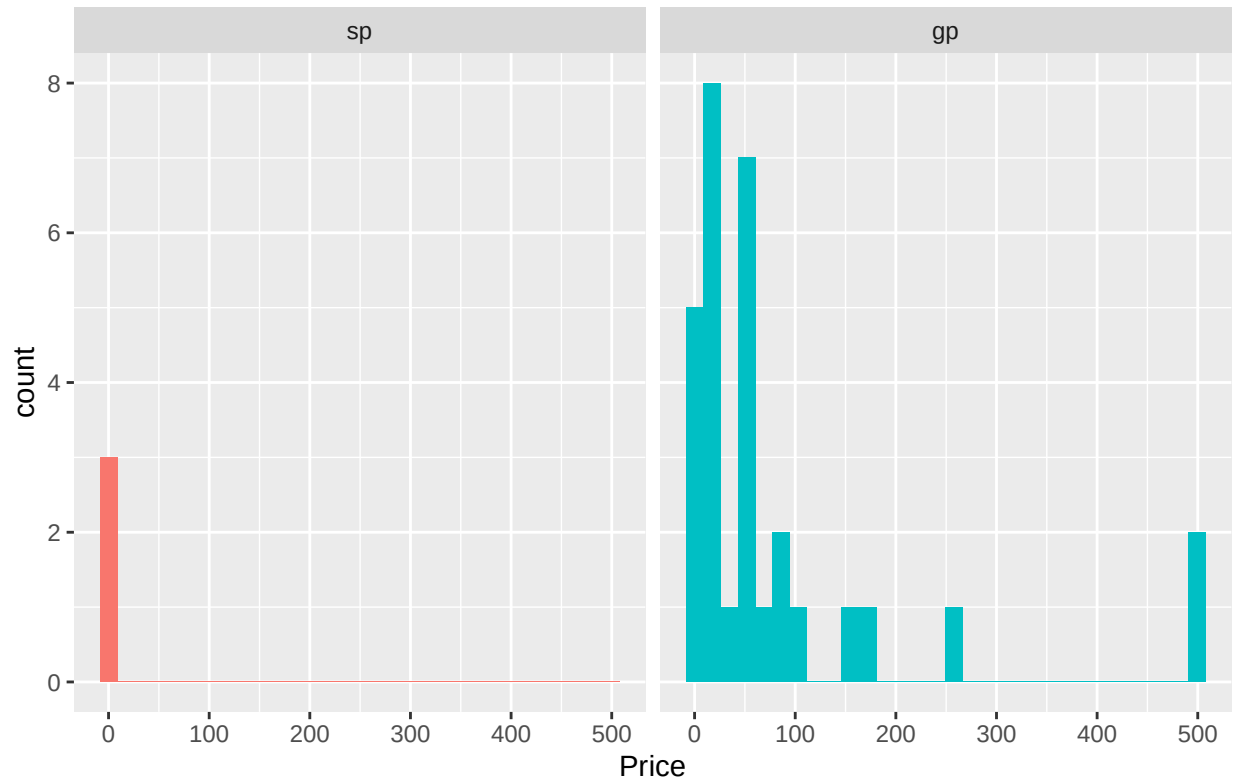
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Clothing Price Distribution



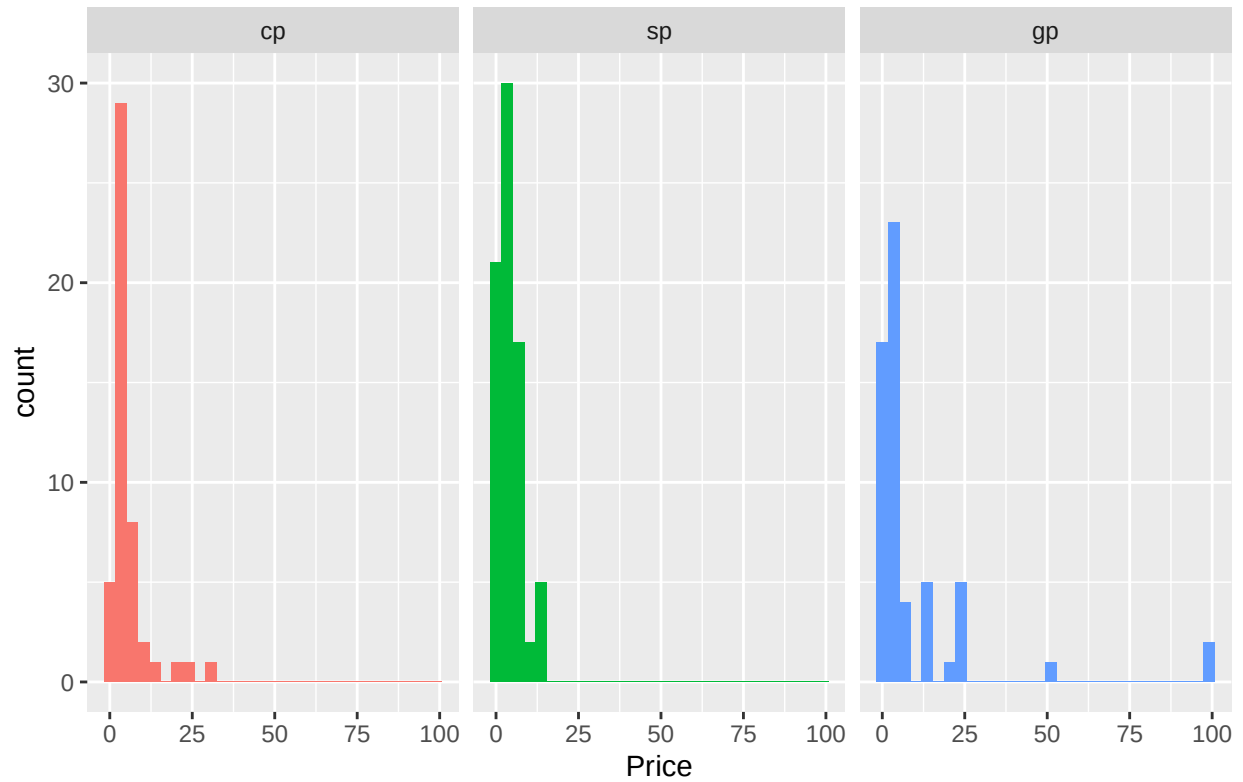
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Entertainment_Gear Price Distribution



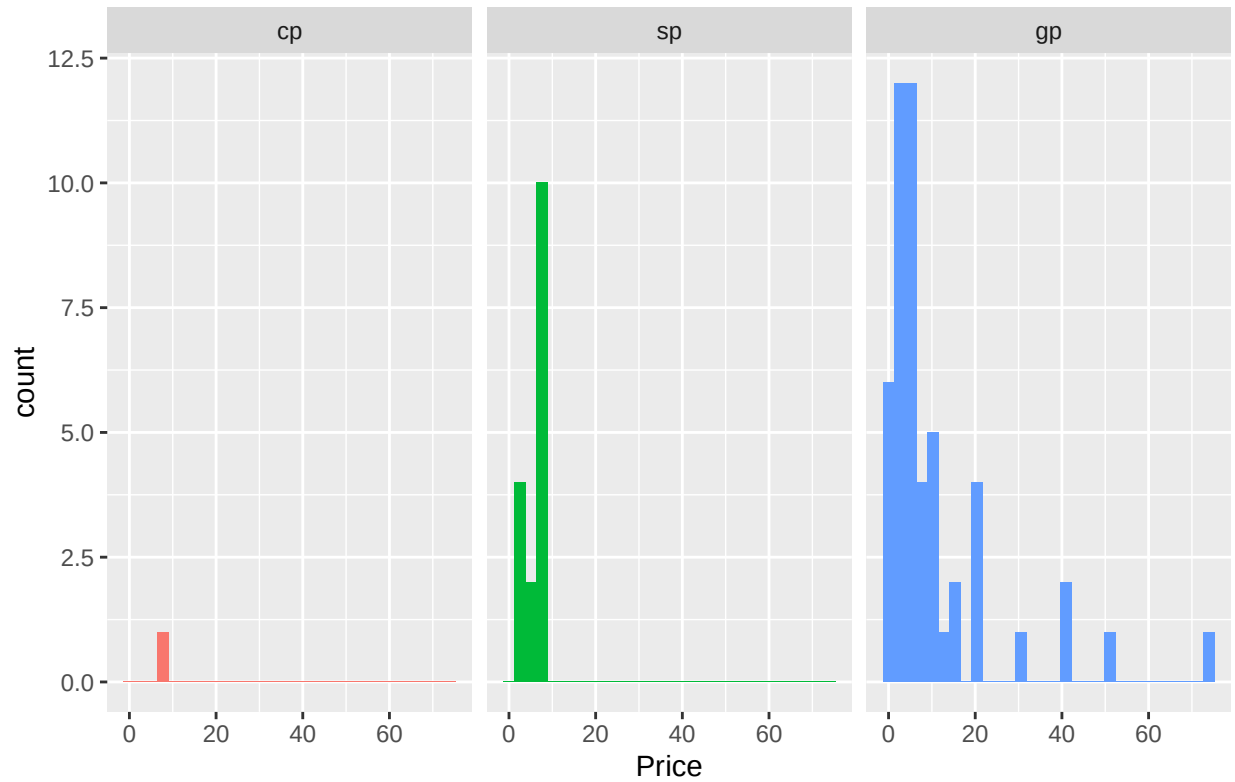
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Foodstuffs Price Distribution



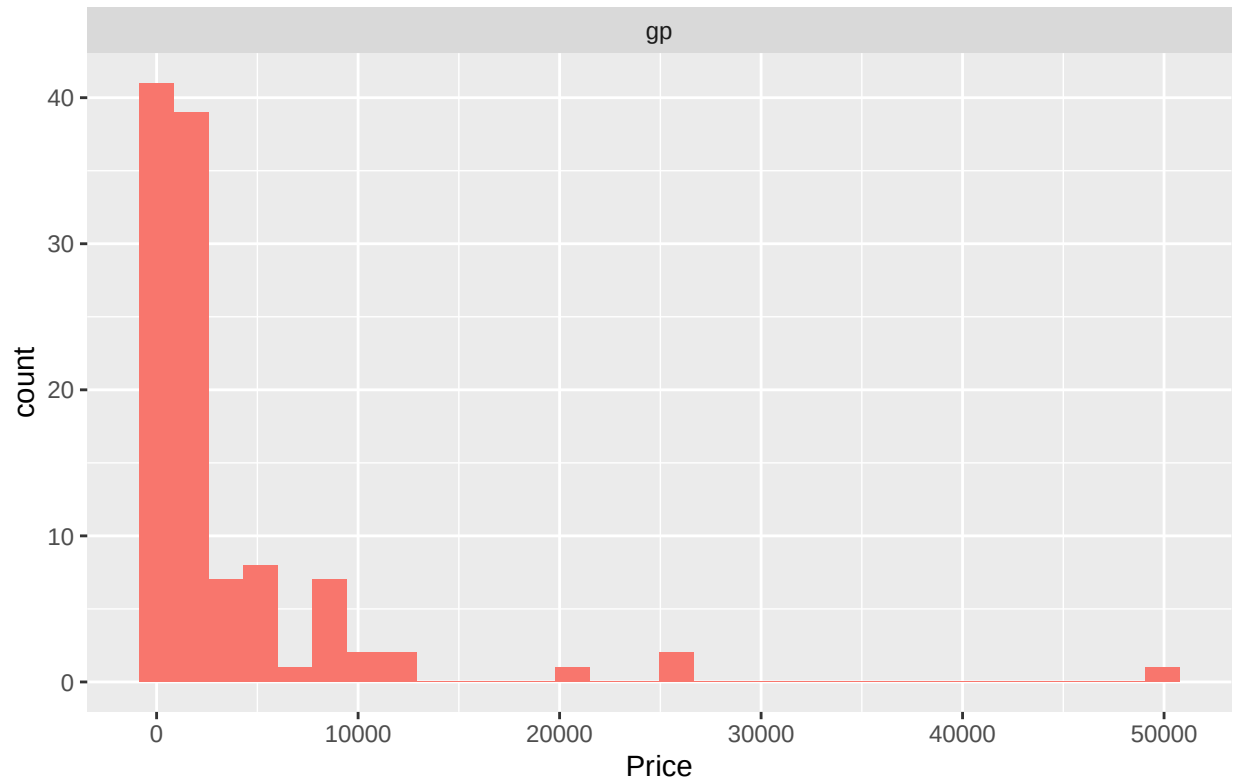
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Furniture Price Distribution



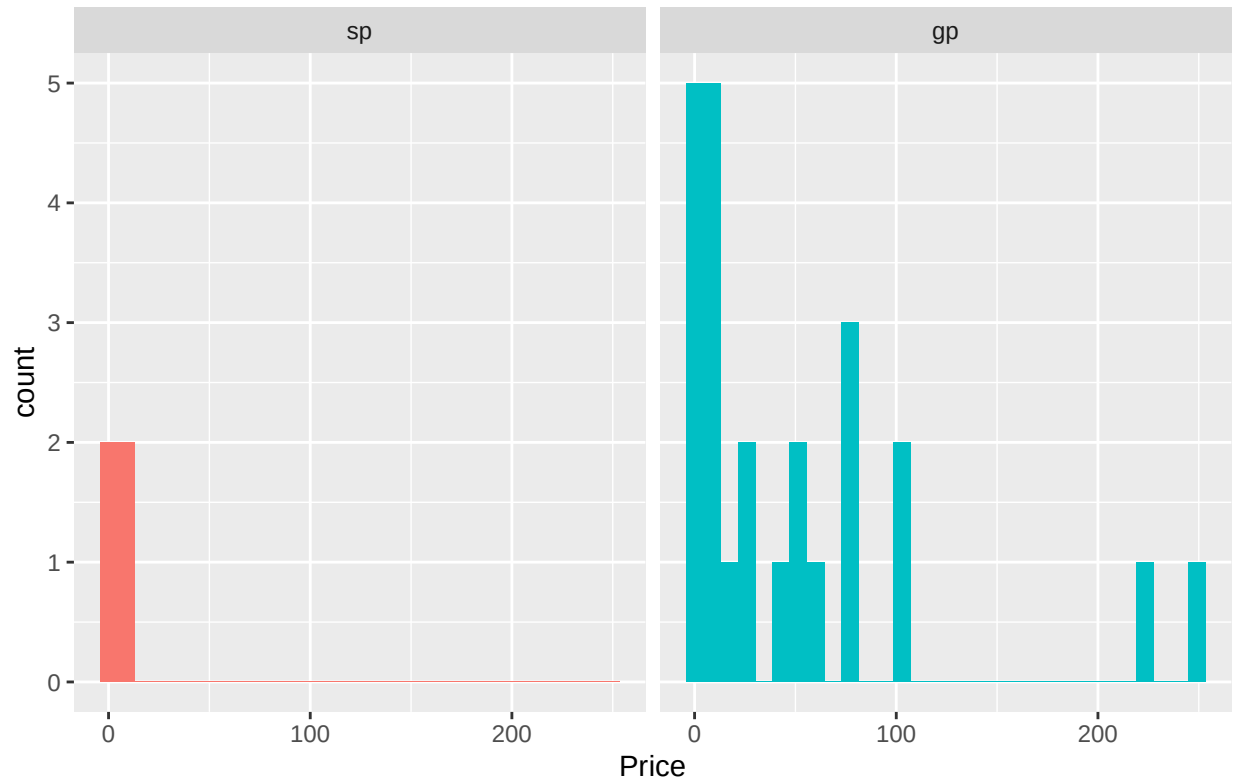
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Land_and_Structures Price Distribution

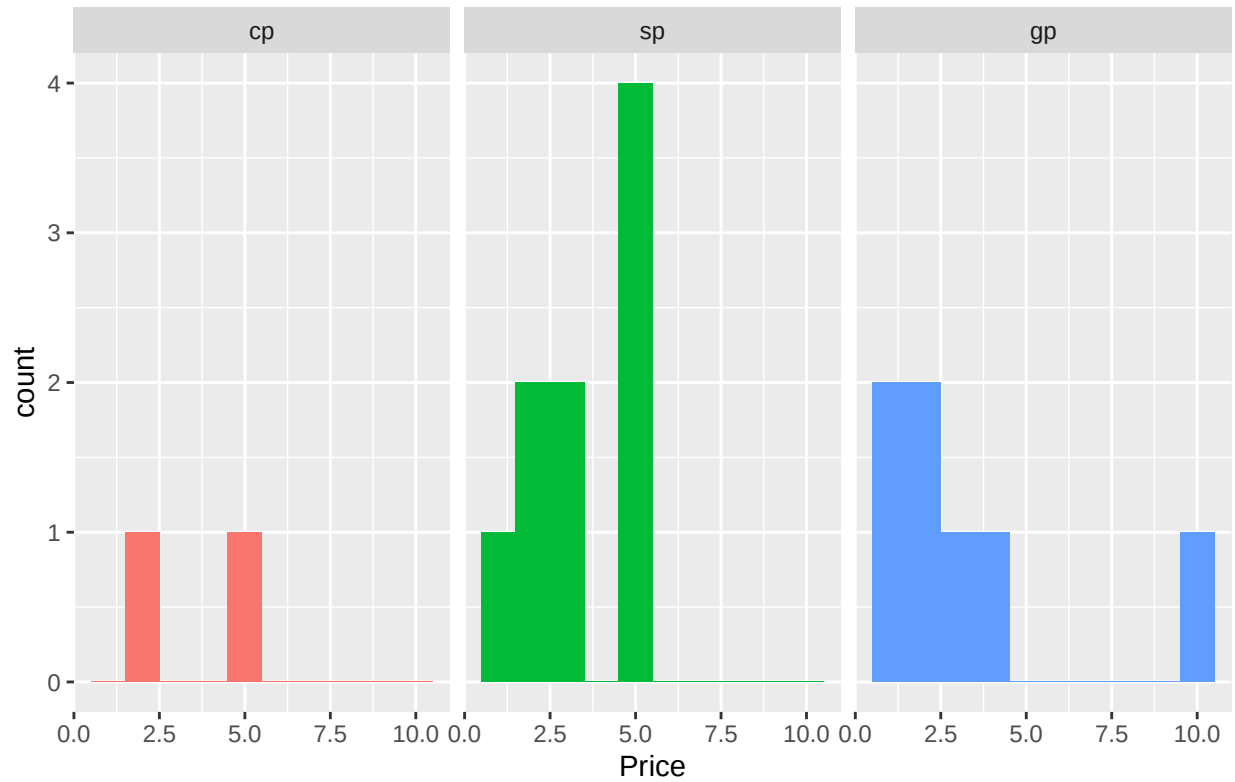


```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Larceny_Gear Price Distribution

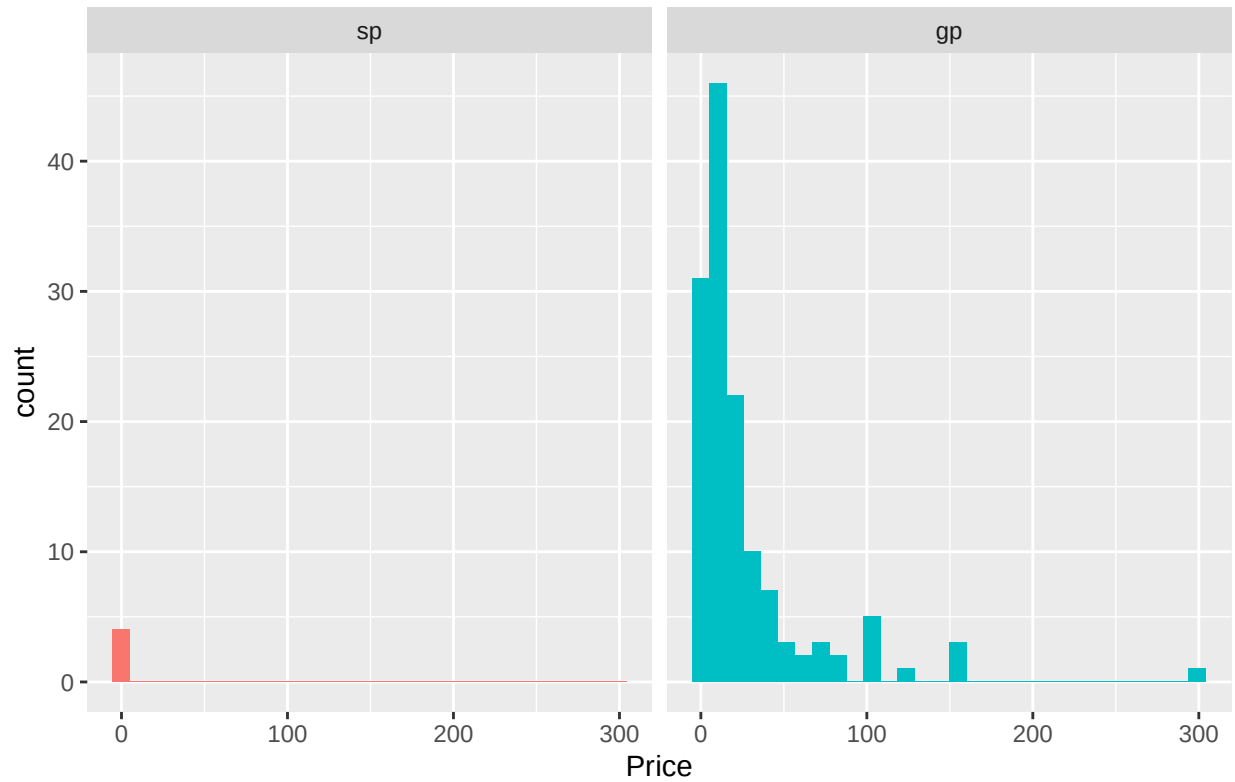


Lodging Price Distribution



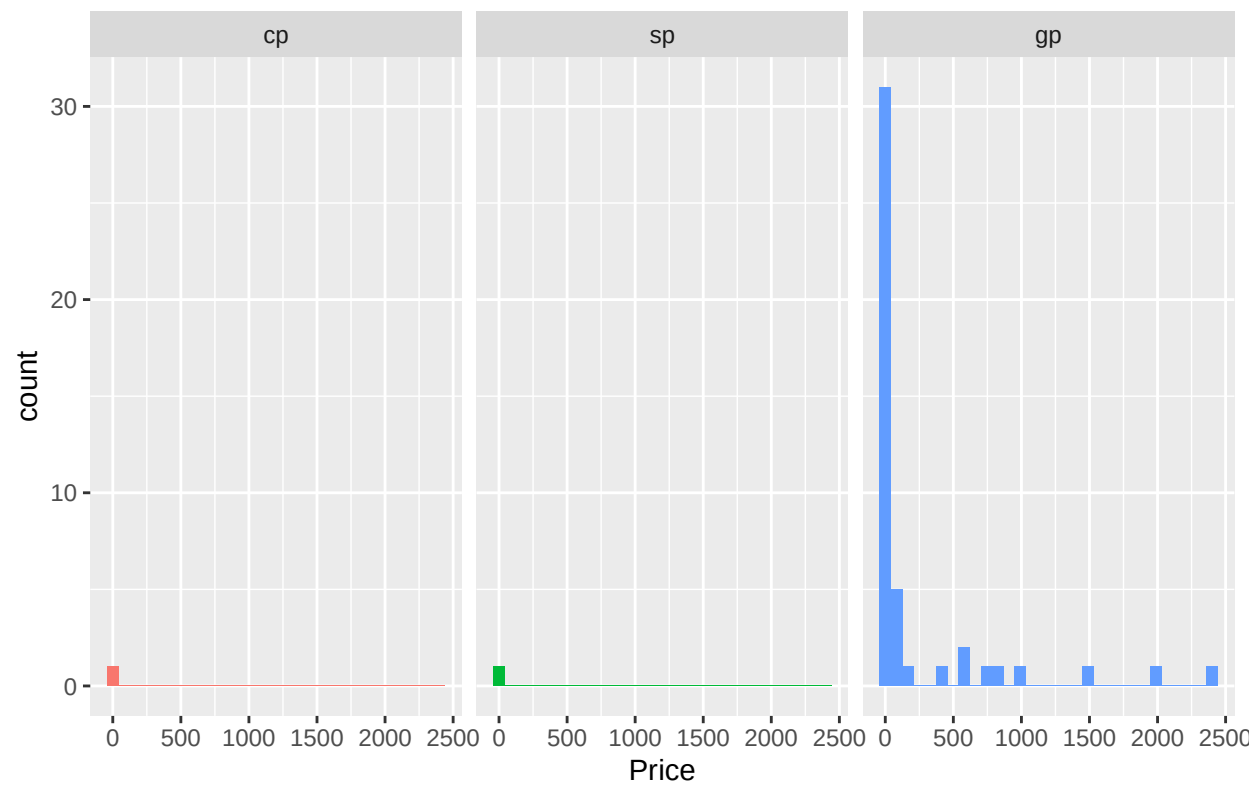
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## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Melee_Weaponry Price Distribution



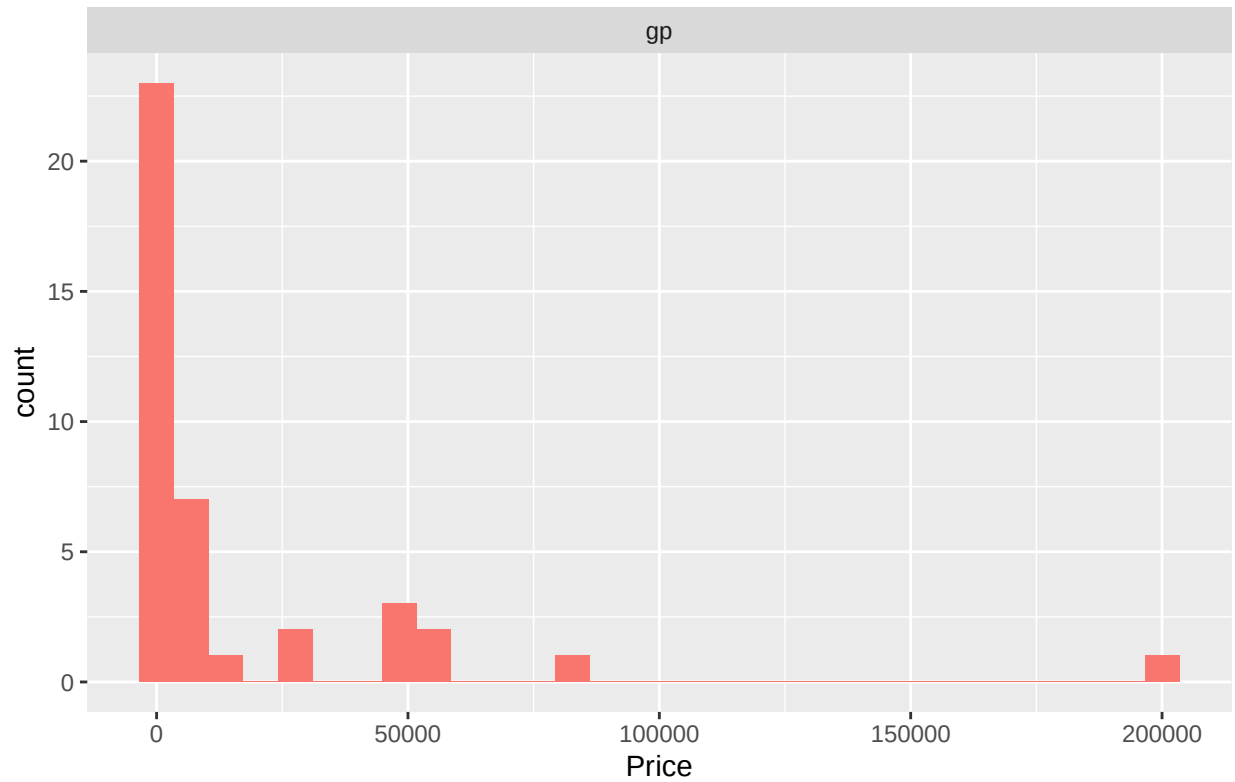
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Mounted_Related_Gear Price Distribution



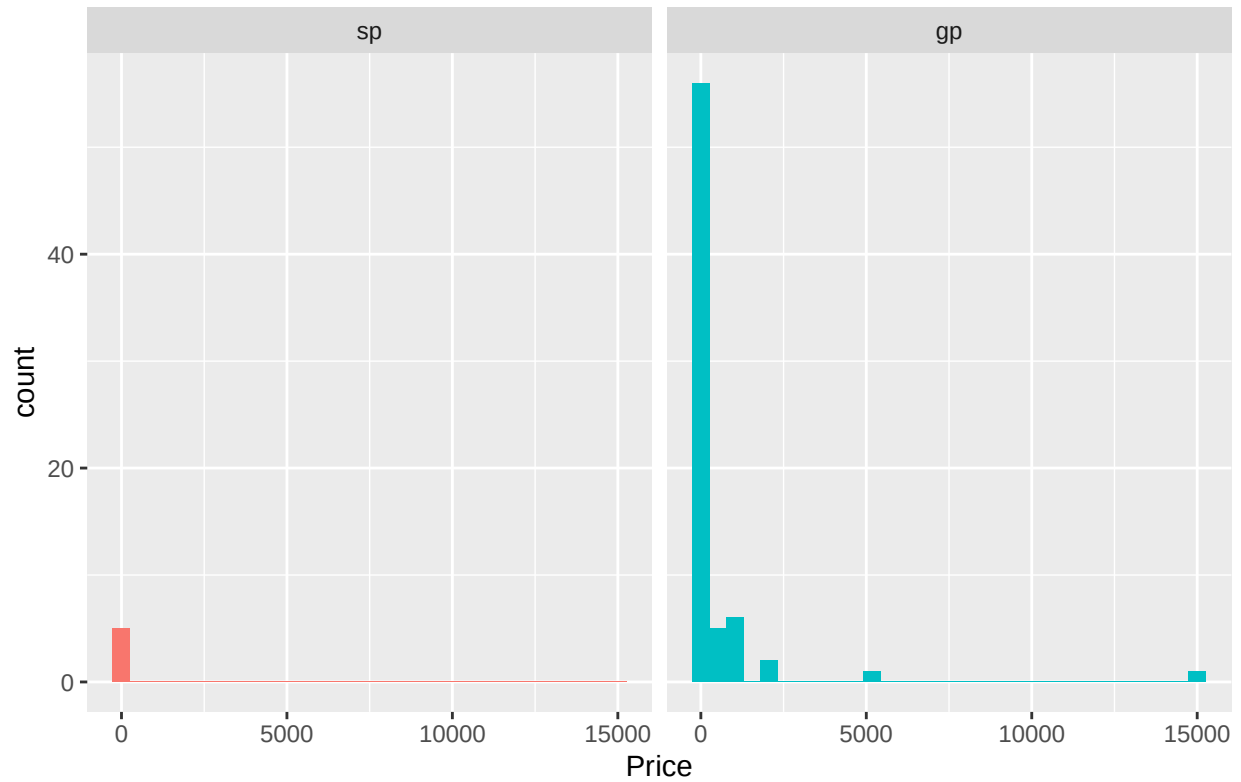
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Naval Price Distribution



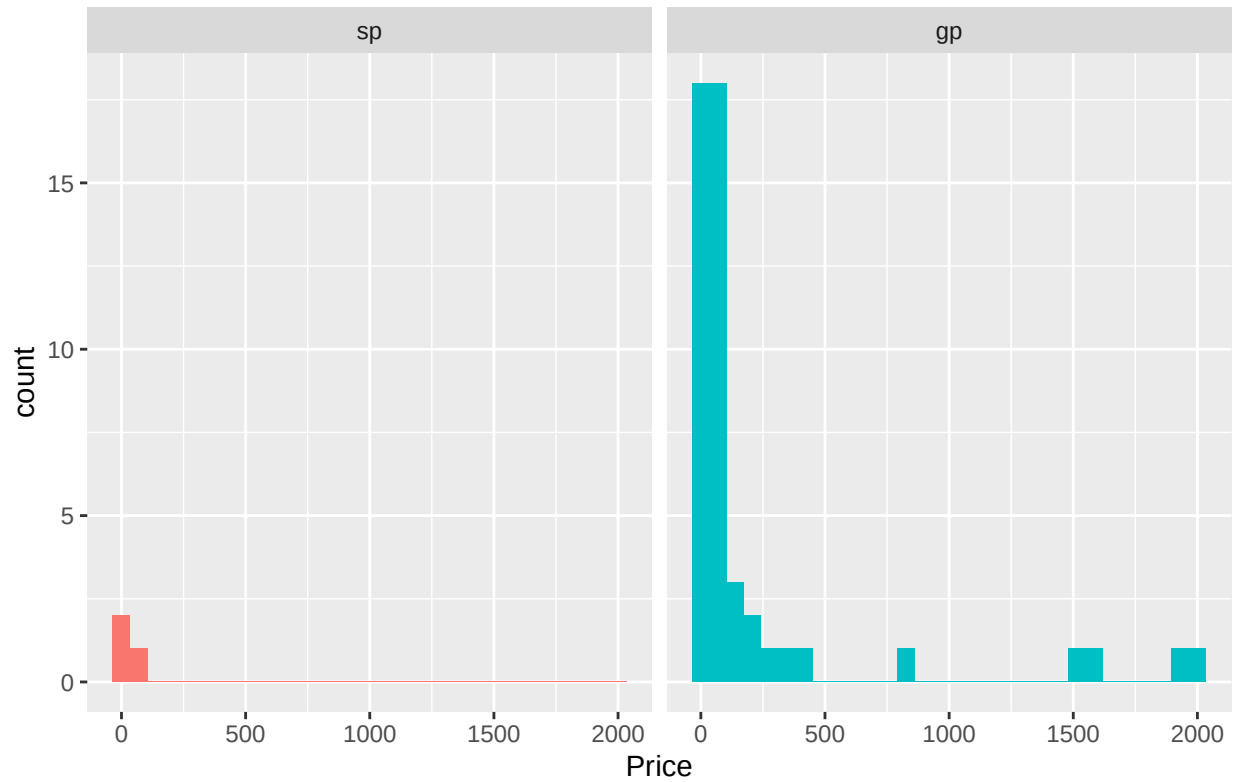
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Ranged_Weaponry Price Distribution



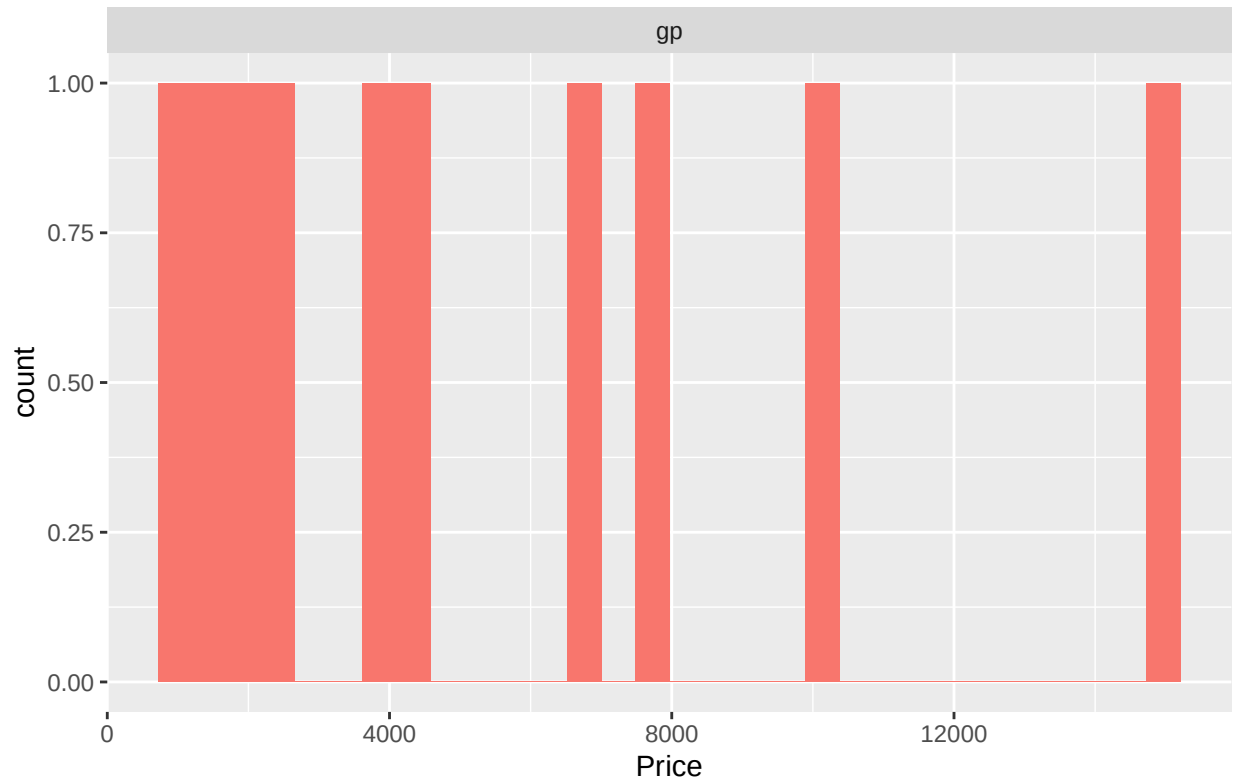
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Religious_Gear Price Distribution



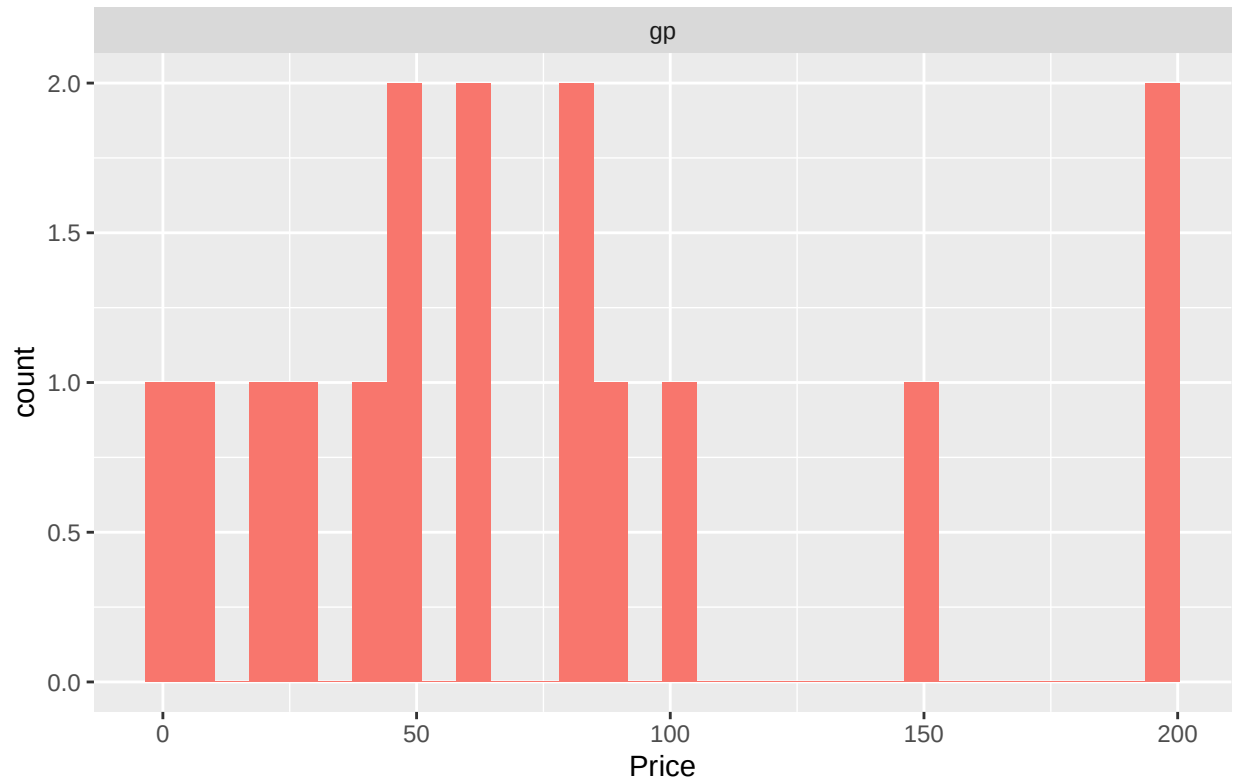
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Slaves Price Distribution



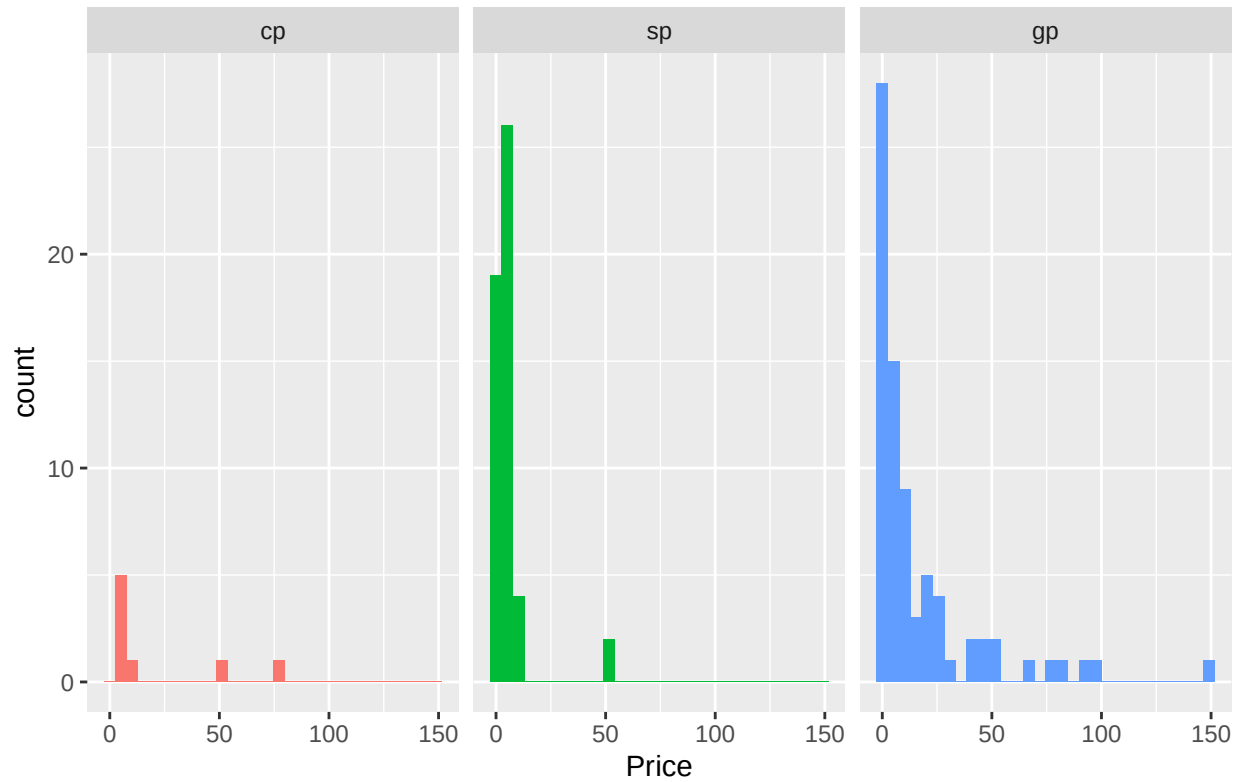
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Subterranean_Items Price Distribution



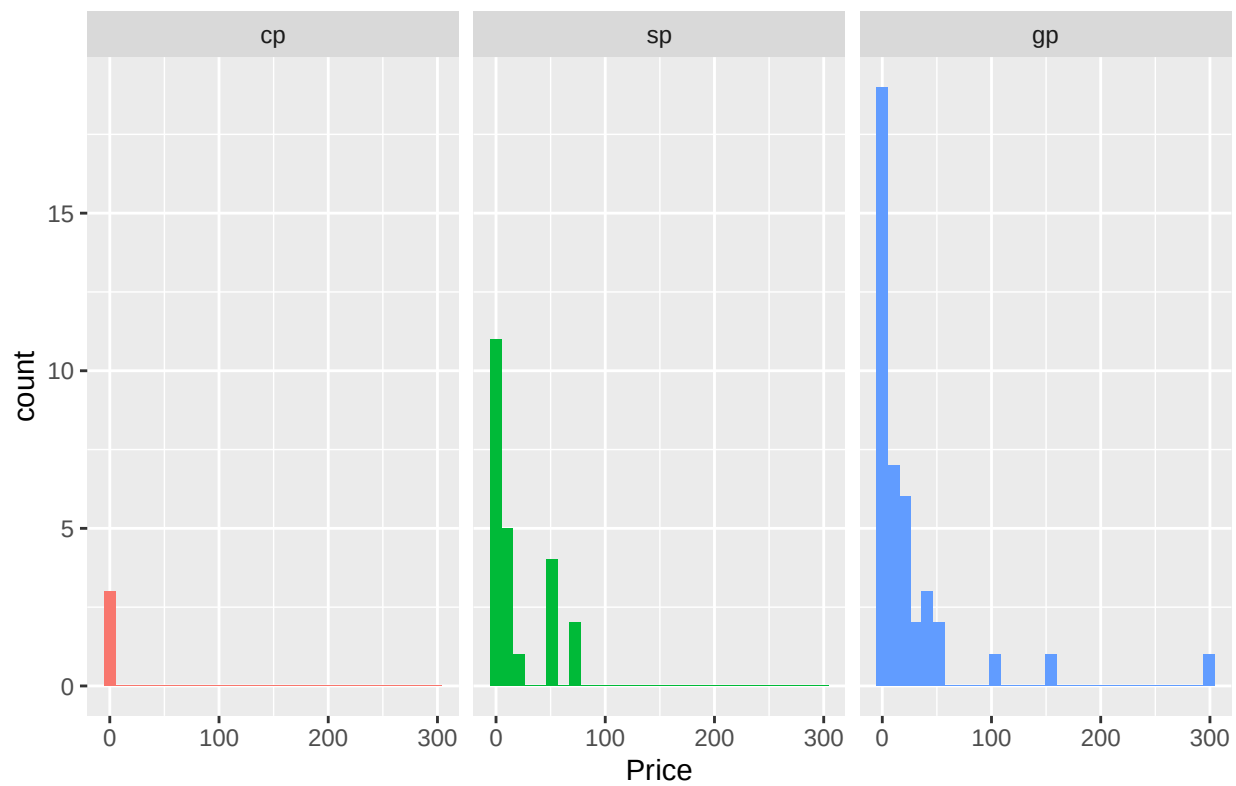
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Tools_and_items Price Distribution



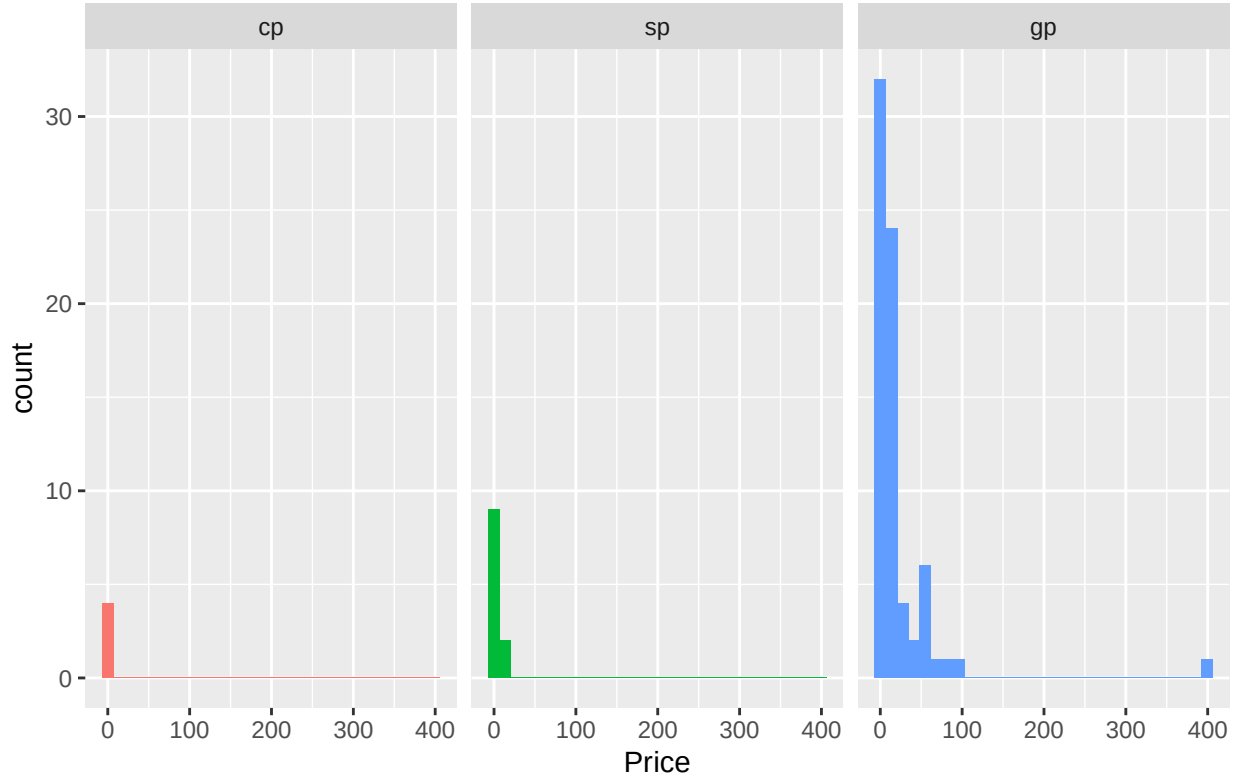
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Trade_Goods Price Distribution



```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Wildreness_Exploration Price Distribution



Conclusion

The current price list is biased toward the use of Gold (gp) over the other currency. Most Categories utilize gp with more frequency than the other currency, with some such as Slaves and Subterranean_Items use gp exclusively. These findings run counter to the intended goal of the price list which is to utilize Copper (cp) as the base standard of cost. The Categories Foodstuffs and Clothing appear to be the closest in line with the stated intent of the price list, but even in these categories there is an overabundance of items which cost gp which suggest these items are not solely luxury goods.

This analysis does not take into consideration modifiers which would likely tip the bias further toward the use of gp. There is evidence that there is crossover between the units in the distributions as well, with items costing in excess of 100 cp still being counted as cp instead of as sp. This may also be true for some items that are in excess of 10 sp and not counted in gp.

A full conclusion cannot be reached with just this initial analysis. It is suggested that additional analysis be carried out on specific categories of the price list to determine the frequency and root cause of the inherent bias observed. It is recommended the next analysis focus on the categories of Foodstuffs and Clothing as these two categories showed the closest resemblance to the desired price distribution.